

OPERATION AND MAINTENANCE MANUAL – PART 6

EXCITATION SYSTEM – HARDWARE AND SYSTEM CONFIGURATION

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Plant Name : VINH-SON HYDRO POWER STATION
Location : VIETNAM
Plant Type : STATIC EXCITATION SYSTEM
Plant Rating : 40MVA

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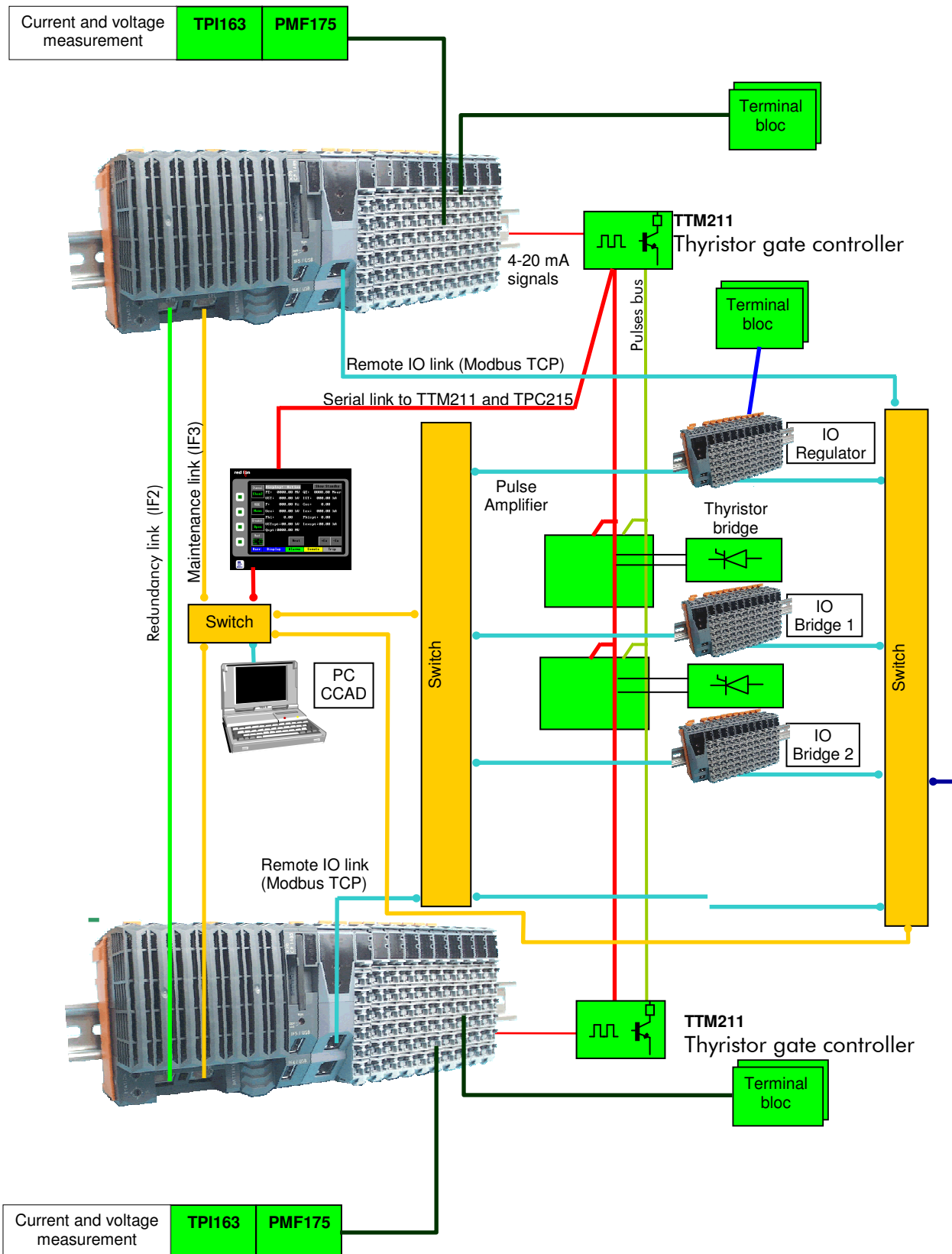
Section 6 **HARDWARE AND SYSTEM CONFIGURATION**

6.1 List of data sheets of the main electronic boards

Name of board	Used	Description of the electronic board
CP1485		Central Processing Unit
IF1082		Ethernet powerlink interface module
X20AO4632		Analog output module, 4 outputs, +/- 10 V, 0.. 20 mA
X20AT4222		Temperatue Module
X20DI9371		Digital input module, 12 inputs, 24VDC
X20AI4632		Analog input module, 4 inputs, +/- 11 V, 0.. 22 mA
TB12		Terminal block 12 pins (for each module)
BC0087		Bus controller field bus interface TCP/IP
PS9400		Supply module for bus controller
ZF0001		Alstom Identification Module
BB80		Bus base for bus controller and supply module
BM11		Bus base for I/O cards
PMF175		Measurement filtering module
TPT214		Thyristor Pulse Transformer
TPI163		Interface for scaling the stator VTs & CTs
LCP		Local control panel
TTM211		Thyristor gate controller board

Note: For details about the electronic boards, refer to the data sheets at the end of this manual.

Regulator architecture arrangement



6.2 Power rectifier

The power converter is based on a disc type thyristor and is assembled in a compact and full wave rectifier bridge. The thyristor (6 thyristors for 3 phases rectifier) are directly tightened between the heat sinks. The cooling of the bridge is performed by an air flow forced through the coolers. A cooling fan is provided for cooling purpose.

The thyristors are protected against over current caused by short circuit. The fuses are fitted as close to the thyristor. Each fuse is fitted with a micro switch to monitor its status.

As a standard, a differential air pressure sensor performs the ventilation monitoring

The bridge is also protected against the over temperature caused by an overload.

The Redundancy Structure is also provided to increase the availability of the equipment.

6.3 Excitation equipment technical data

The specification is included in Part 8.